PyGameMaker — Tutorial 7: Platformer — Teacher Guide

Companion for the in-app tutorial (**Help > Tutorials > Platformer: Run, Jump, Collect**, 4 pages) and the matching student handout and worksheet. Students should have finished Tutorials 2 and 6 (events, collisions, solid objects, score).

Overview

Students build a platformer in three phases: gravity, running and jumping; coins, spikes and a flag; and a score and lives display. New ideas: **gravity**, **jumping** with a vertical-speed impulse, **hazards**, and lives that survive a room restart.

> Gravity is a good place to ask "what does the computer do every step?": add 0.5 to the vertical speed, then move. A jump is just a big upward speed that gravity slowly cancels.

Suggested Timing (60 minutes)

The tutorial says 25-30 minutes for a confident student; level building adds time.

Segment	Time	What happens
Recap and predict	5 min	Ask what forces act on a jumping character
Phase 1: gravity, running, jumping	15-20 min	Pages 1-2; the level layout takes time
Phase 2: coins, spikes, flag	15 min	Page 3
Phase 3: controller and lives	10 min	Page 4
Play and tune	5 min	Change gravity and jump speed; play each other's levels
Worksheet / exit ticket	5 min	Worksheet parts A-C

For a 45-minute slot, do Phases 1-2 and leave Phase 3 for homework.

Step-by-Step Walkthrough and Common Issues

Phase 1: gravity, running, jumping

- Gravity is set once in the player's **Create** event: direction **270** (down), strength **0.5**. Angles start at 0 (right) and go counter-clockwise; 90 is up.
- **No key** must be *Set horizontal speed to 0*, **not** *Stop movement*. Stop Movement also wipes the vertical speed that gravity builds, so the player hangs in the air (in the reference project it drifts about 15 pixels in half a second instead of falling more than 100). This is the most common cause of "the player does not fall".
- The jump is on **Key press** (once per press), not *Keyboard (held)*.
- The ground needs to be **Solid**, and the player needs **Collision with obj_ground** with *Stop movement*; otherwise it falls through.
- **Keep the space above the player free** when placing them. A platform directly above the starting position blocks the jump and looks like "the jump does not work". The jump in the reference project is about 95 pixels (three tiles) high.

- Players must be placed standing on the ground (or above it). Snap to Grid 32×32 makes that easy.

Phase 2: coins, hazards and flag

- All the interactions are collision events **on the player**; the coin, spike and flag objects need no events.
- The spike uses *Restart room*. In Phase 2 nothing is remembered between restarts; the coins come back.
- The flag shows a message; there is no next level yet.

Phase 3: game controller and lives

- The controller must be placed in the room.
- **Set the score and lives in Game Start, not Create.** A Create event runs again every time the room restarts, and a spike restarts the room, so with Create the lives were reset to 3 after every spike and could never run out. (The tutorial used Create until this was corrected; if a student's screen shows "Create", it is an old copy of the instructions.) Game Start runs once, at the very start of the game.
- With Game Start, the score is also kept across restarts, so a student can collect the same coins again after each death. Ask students whether that is fair; resetting the score in the spike event is a good change.
- Nothing special happens when the lives reach 0 (no Game Over): that is a challenge at the end of the tutorial.

> **Reference projects.** Download the checkpoint projects from the wiki (07_platformer_checkpoints.zip): one project for the end of each phase, with a small level. Give a stuck student the previous phase.

Vocabulary Introduced

Term	What it means here
Gravity	Adds to the vertical speed every step, in the given direction
Vertical speed	Up is negative, down is positive
Impulse	A single change of speed, like the jump
Hazard	An object that harms the player (spike)
Game Start	An event that runs once when the game begins, not on room restarts

Discussion Questions

- Why is the jump speed negative?
- What happens to the jump if we double the gravity? Half the jump speed?
- Why does the player need a collision event with the ground as well as the ground being Solid?
- Why can't the lives be set in Create?

Differentiation

- **Support:** give the Phase 1 checkpoint and let students build the level on top; keep the level small.
- **Extension:** moving platforms; patrolling enemies; a double jump; Game Over at 0 lives; a second room from the flag.

Worksheet Answer Key

Part A: 1-C, 2-D, 3-A, 4-B.

Part B: 1. Create (Set gravity). 2. Key press: Up Arrow. 3. Collision with obj_coin (on the player). 4. Game Start (on the controller). 5. Draw (on the controller).

Part C:

1. About 20 steps (10 ÷ 0.5 = 20).
2. *Stop movement* in **No key**; it should be *Set horizontal speed to 0*.
3. The room restarts and the Create event runs again, so the lives are set back to 3 after every spike and never run out. It has to be Game Start.

Parts D and E: completion and reflection.

Rubric: Platformer

Level	What the game shows
4 - Complete	Gravity, running and one jump per press work; platforms are solid; coins score and disappear; spikes cost a life that stays lost; flag shows a message; score and lives are displayed
3 - Working	Movement, jumping, coins and spikes work; lives or display is missing
2 - Partly there	The player falls, runs and jumps, but coins or hazards do not work
1 - Started	The player and ground exist, but the player falls through or cannot move

End-of-Session Checklist

- [ ] Every student has a saved project (Ctrl+S)
- [ ] Every student reached at least Phase 1 (run and jump)
- [ ] Note who needs the checkpoint project next time